

For the use of a Registered Medical Practitioner only
PRESCRIBING INFORMATION

IRBEL TABLETS

(Irbesartan Tablets)

COMPOSITION

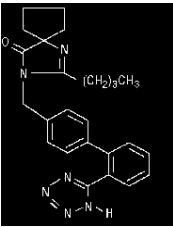
Irbel Tablets 150 mg
Each film-coated tablet contains:
Irbesartan Ph. Eur. 150 mg

Irbel Tablets 300 mg
Each film-coated tablet contains:
Irbesartan Ph. Eur. 300 mg

Inactive ingredients:
Lactose, Microcrystalline Cellulose, Croscarmellose Sodium, Hydroxypropyl Methylcellulose, Colloidal Anhydrous Silica, Magnesium Stearate, Opadry II white OY-LS-28900 (Lactose, Hypromellose, Titanium dioxide, Macrogol), Purified Water

DESCRIPTION

IRBEL tablets contains irbesartan, which is an angiotensin II receptor (AT1 subtype) antagonist. Irbesartan is a non-peptide compound, chemically described as a 2-butyl-3-[p-(o-1H-tetrazol-5-ylphenyl)benzyl]-1,3-diazaspiro[4.4]non-1-en-4-one. Its empirical formula is C₂₅H₂₈N₆O and molecular weight is 428.5, and the structural formula:



Product Description:

For 150 mg strength: White to off white, film coated oval shaped, biconvex tablet debossed with '14' on one side and plain on the other.

For 300 mg strength: White to off white, film coated oval shaped, biconvex tablet debossed with '15' on one side and plain on the other.

PHARMACODYNAMIC AND PHARMACOKINETIC PROPERTIES

Pharmacodynamics

Mechanism of Action

Irbesartan is a potent, orally active, selective angiotensin-II receptor (type AT1) antagonist. It is expected to block all actions of angiotensin-II mediated by the AT1 receptor, regardless of the source or route of synthesis of angiotensin-II. The selective antagonism of the angiotensin-II (AT1) receptors results in increases in plasma renin levels and angiotensin-II levels, and a decrease in plasma aldosterone concentration. Serum potassium levels are not significantly affected by irbesartan alone at the recommended doses. Irbesartan does not inhibit ACE (kininase-II), an enzyme which generates angiotensin-II and also degrades bradykinin into inactive metabolites. Irbesartan does not require metabolic activation for its activity.

Pharmacokinetics

After oral administration, irbesartan is well absorbed. The absolute bioavailability values have been reported to be approximately 60-80%. Concomitant food intake does not significantly influence the bioavailability of irbesartan.

Irbesartan exhibits linear and dose proportional pharmacokinetics over the dose range of 10 to 600 mg. A less than proportional increase in oral absorption at doses beyond 600 mg (twice the maximal recommended dose) has been reported; the mechanism for this is unknown. Peak plasma concentrations are attained at 1.5 - 2 hours after oral administration.

Plasma protein binding is approximately 96%, with negligible binding to cellular blood components. The volume of distribution is 53 - 93 liters. Following oral or intravenous administration of 14C irbesartan, 80 - 85% of the circulating plasma radioactivity is attributable to unchanged irbesartan.

Irbesartan is metabolised by the liver via glucuronide conjugation and oxidation. The major circulating metabolite is irbesartan glucuronide (approximately 6%). Irbesartan is primarily oxidised by the cytochrome P450 enzyme CYP2C9; isoenzyme CYP3A4 has negligible effect.

Irbesartan and its metabolites are eliminated by both biliary and renal pathways. After either oral or IV administration of 14C irbesartan, about 20% of the radioactivity is recovered in the urine, and the remainder in the faeces. Less than 2% of the dose is excreted in the urine as unchanged irbesartan.

The total body and renal clearance are 157 - 176 and 3 - 3.5 ml/min, respectively. The terminal elimination half-life of irbesartan is 11 - 15 hours. Steady-state plasma concentrations are attained within 3 days after initiation of a once daily dosing regimen. Limited accumulation of irbesartan (< 20%) has been reported in plasma upon repeated once-daily dosing.

Pharmacokinetics in special populations

Gender

Higher plasma concentrations of irbesartan are reported in female hypertensive patients. However, there was no difference in the half-life and accumulation of irbesartan. No dosage adjustment is necessary in female patients.

Geriatric

Irbesartan AUC and Cmax values were also somewhat greater in elderly subjects (≥ 65 years) than those of young subjects (18 - 40 years). However the terminal half-life was not significantly altered. No dosage adjustment is necessary in elderly patients.

Renal impairment

In patients with renal impairment or those undergoing haemodialysis, the pharmacokinetic parameters of irbesartan are not significantly altered. Irbesartan is not removed by haemodialysis.

Hepatic impairment

In patients with mild to moderate cirrhosis, the pharmacokinetic parameters of irbesartan are not significantly altered. No information is available on pharmacokinetics of irbesartan in patients with severe hepatic impairment.

Pediatric patients

It has been reported that in hypertensive children receiving single and multiple daily doses of irbesartan (2 mg/kg) up to maximum daily dose of 150 mg showed comparable Cmax, AUC and clearance rates to those reported in adult patients receiving 150 mg irbesartan daily. A limited accumulation of irbesartan (18%) in plasma has been reported upon repeated once daily dosing.

INDICATIONS

Y Treatment of essential hypertension.

Y Treatment of diabetic nephropathy with an elevated serum creatinine and proteinuria (> 300 mg/day) in patients with type 2 diabetes and hypertension. In this population, irbesartan reduces the rate of progression of neuropathy as measured by the occurrence of doubling of serum creatinine or end stage renal disease (need for dialysis or renal transplantation).

DOSE AND METHOD OF ADMINISTRATION

The usual recommended initial and maintenance dose is 150 mg once daily, with or without food. Irbesartan at a dose of 150 mg once daily generally provides a better 24 hour blood pressure control than 75 mg. However, initiation of therapy with 75 mg could be considered, particularly in haemodialysis patients and in the elderly over 75 years.

In patients insufficiently controlled with 150 mg once daily, the dose of irbesartan can be increased to 300 mg, or other anti-hypertensive agents can be added. In particular, the addition of a diuretic such as hydrochlorothiazide has been shown to have an additive effect with irbesartan.

In hypertensive type 2 diabetic patients, therapy should be initiated at 150 mg once daily and titrated up to 300 mg once daily as the preferred maintenance dose for treatment of renal disease.

Renal benefit of irbesartan has been reported in hypertensive type 2 diabetic patients where irbesartan was used in addition to other antihypertensive agents, as needed, to reach target blood pressure.

Special Populations

Renal impairment: No dosage adjustment is necessary in patients with impaired renal function. A lower starting dose (75 mg) should be considered for patients undergoing haemodialysis.

Hepatic impairment: No dosage adjustment is necessary in patients with mild to moderate hepatic impairment. There is no clinical experience in patients with severe hepatic impairment.

Elderly patients: Although consideration should be given to initiating therapy with 75 mg in patients over 75 years of age, dosage adjustment is not usually necessary for the elderly.

Paediatric patients: Safety and efficacy of irbesartan has not been established in children.

Intravascular volume depletion: Volume and/or sodium depletion should be corrected prior to administration of irbesartan.

CONTRAINDICATIONS

Y Hypersensitivity to the active substance, or to any of the excipients.
Y Second and third trimester of pregnancy.
Y Lactation

WARNINGS AND PRECAUTIONS

Intravascular volume depletion: Symptomatic hypotension, especially after the first dose, may occur in patients who are volume and/or sodium depleted by vigorous diuretic therapy, dietary salt restriction, diarrhoea or vomiting. Such conditions should be corrected before the administration of irbesartan.

Renovascular hypertension: There is an increased risk of severe hypotension and renal insufficiency when patients with bilateral renal artery stenosis or stenosis of the artery to a single functioning kidney are treated with medicinal products that affect the renin-angiotensin-aldosterone system. While this is not documented with irbesartan, a similar effect should be anticipated with angiotensin-II receptor antagonists.

Renal impairment and kidney transplantation: When irbesartan is used in patients with impaired renal function, a periodic monitoring of potassium and creatinine serum levels is recommended. There is no experience regarding the administration of irbesartan in patients with recent kidney transplantation. In patients whose renal function may depend on the activity of the renin-angiotensin-aldosterone system (e.g. patients with severe congestive heart failure), treatment with angiotensin-converting enzyme inhibitors has been associated with oliguria and/or progressive azotemia and (rarely) with acute renal failure and/or death. Irbesartan would be expected to behave similarly.

Hypertensive patients with type 2 diabetes and renal disease: The effects of irbesartan both on renal and cardiovascular events were not uniform across all subgroups, in patients with advanced renal disease. In particular, they appeared less favorable in women and non-white subjects.

Hyperkalaemia: As with other medicinal products that affect the renin-angiotensin-aldosterone system, hyperkalaemia may occur during the treatment with irbesartan, especially in the presence of renal impairment, overt proteinuria due to diabetic renal disease, and/or heart failure. Close monitoring of serum potassium in patients at risk is recommended (see **DRUG INTERACTIONS**).

Lithium: The combination of lithium and irbesartan is not recommended (see **DRUG INTERACTIONS**).

Aortic and mitral valve stenosis, obstructive hypertrophic cardiomyopathy: As with other vasodilators, special caution is indicated in patients suffering from aortic or mitral stenosis, or obstructive hypertrophic cardiomyopathy.

Primary aldosteronism: Patients with primary aldosteronism generally will not respond to anti-hypertensive medicinal products acting through inhibition of the renin-angiotensin system. Therefore, the use of irbesartan is not recommended.

General: In patients whose vascular tone and renal function depend predominantly on the activity of the renin-angiotensin-aldosterone system (e.g. patients with severe congestive heart failure or underlying renal disease, including renal artery stenosis), treatment with angiotensin converting enzyme inhibitors or angiotensin-II receptor antagonists that affect this system has been associated with acute hypotension, azotaemia, oliguria, or rarely acute renal failure. As with any anti-hypertensive agent, excessive blood pressure decrease in patients with ischaemic cardiopathy or ischaemic cardiovascular disease could result in a myocardial infarction or stroke.

As observed for angiotensin converting enzyme inhibitors, irbesartan and the other angiotensin antagonists are apparently less effective in lowering blood pressure in black people than in non-blacks, possibly because of higher prevalence of low-renin states in the black hypertensive population.

Pregnancy: Angiotensin II Receptor Antagonists (AIIRAs) should not be initiated during pregnancy. Unless continued AIIRA therapy is considered essential, patients planning pregnancy should be changed to alternative anti-hypertensive treatments which have an established safety profile for use in pregnancy. When pregnancy is diagnosed, treatment with AIIRAs should be stopped immediately, and, if appropriate, alternative therapy should be started (see **CONTRAINDICATIONS, USE IN SPECIAL POPULATIONS**).

This medicinal product contains lactose. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicinal product.

Effects on ability to drive and use machines: No information is available on the effects on the ability to drive and use machines. Based on its pharmacodynamic properties, irbesartan is unlikely to affect this ability. When driving vehicles or operating machines, it should be taken into account that dizziness or weariness may occur during treatment.

DRUG INTERACTIONS

Diuretics and other antihypertensive agents: Other antihypertensive agents may increase the hypotensive effects of irbesartan; however irbesartan has been safely administered with other antihypertensive agents, such as beta-blockers, long-acting calcium channel blockers, and thiazide diuretics. Prior treatment with high dose diuretics may result in volume depletion and a risk of hypotension when initiating therapy with irbesartan.

Potassium supplements and potassium-sparing diuretics: Based on experience with the use of other medicinal products that affect the renin-angiotensin system, concomitant use of potassium-sparing diuretics, potassium supplements, salt substitutes containing potassium or other medicinal products that may increase serum potassium levels (e.g. heparin) may lead to increases in serum potassium and is, therefore, not recommended.

Lithium:

Reversible increases in serum lithium concentrations and toxicity have been reported during concomitant administration of lithium with angiotensin converting enzyme inhibitors. Similar effects have been very rarely reported with irbesartan so far. Therefore, this combination is not recommended. If the combination proves necessary, careful monitoring of serum lithium levels is recommended.

Non-steroidal anti-inflammatory drugs:

When angiotensin II antagonists are administered simultaneously with non-steroidal anti-inflammatory drugs (i.e. selective COX-2 inhibitors, acetylsalicylic acid (> 3g/day) and non-selective NSAIDs), attenuation of the antihypertensive effect may occur.

As with ACE inhibitors, concomitant use of angiotensin II antagonists and NSAIDs may lead to an increased risk of worsening of renal function, including possible acute renal failure, and an increase in serum potassium, especially in patients with poor pre-existing renal function. The combination should be administered with caution, especially in the elderly. Patients should be adequately hydrated and consideration should be given to monitoring renal function after initiation of concomitant therapy, and periodically thereafter.

Additional information on irbesartan interactions:

Irbesartan is mainly metabolised by CYP2C9 and to a lesser extent by glucuronidation. No significant pharmacokinetic or pharmacodynamic interactions have been reported with co-administration of irbesartan and warfarin (a medicinal product metabolised by CYP2C9). The effects of CYP2C9 inducers such as rifampicin on the pharmacokinetic of irbesartan have not been reported. The pharmacokinetic of digoxin was not altered by co-administration of irbesartan.

SIDE EFFECTS/ADVERSE REACTIONS

The following list presents the adverse drug reactions that have been reported in hypertensive patients receiving irbesartan.

- **Immune system disorders:**
Not known: hypersensitivity reactions such as angioedema, rash, urticaria
- **Metabolism and nutrition disorders:**
Not known: hyperkalaemia
- **Nervous system disorders:**
Common: dizziness, orthostatic dizziness*
Not known: vertigo, headache
- **Ear and labyrinth disorder:**
Not known: tinnitus
- **Cardiac disorders:**
Uncommon: tachycardia
- **Vascular disorders:**
Common: orthostatic hypotension*
Uncommon: flushing
- **Respiratory, thoracic and mediastinal disorders:**
Uncommon: cough
- **Gastrointestinal disorders:**
Common: nausea/vomiting
Uncommon: diarrhoea, dyspepsia/heartburn
Not known: dysgeusia
- **Hepatobiliary disorders:**
Uncommon: jaundice
Not known: hepatitis, abnormal liver function
- **Skin and subcutaneous tissue disorders:**
Not known: leukocytoclastic vasculitis
- **Musculoskeletal and connective tissue disorders:**
Common: musculoskeletal pain*
Not known: arthralgia, myalgia (in some cases associated with increased plasma creatine kinase levels), muscle cramps
- **Renal and urinary disorders:**
Not known: impaired renal function including cases of renal failure in patients at risk
- **Reproductive system and breast disorders:**
Uncommon: sexual dysfunction
- **General disorders and administration site conditions:**
Common: fatigue
Uncommon: chest pain
- **Investigations:**
Very common: Hyperkalaemia* occurred more often in diabetic patients receiving irbesartan. In diabetic hypertensive patients with microalbuminuria and normal renal function, hyperkalaemia (≥ 5.5 mEq/L) reported with irbesartan 300 mg. In diabetic hypertensive patients with chronic renal insufficiency and overt proteinuria, hyperkalaemia (≥ 5.5 mEq/L) reported with irbesartan.

Common: significant increases in plasma creatine kinase were commonly reported in patients receiving irbesartan. None of these increases were associated with identifiable clinical musculoskeletal events. In hypertensive patients with advanced diabetic renal disease receiving irbesartan, a decrease in haemoglobin*, which was not clinically significant, has been reported.

Terms marked with a star (*) refer to the adverse reactions that have been additionally reported in diabetic hypertensive patients with chronic renal insufficiency and overt proteinuria.

Pediatric patients: Following are the adverse events reported in hypertensive children and adolescents aged 6 to 16 years: headache, hypotension, dizziness, cough.. It has been reported that the most frequent laboratory abnormalities observed were creatinine increases and elevated CK values in child recipients.

Additionally following adverse events have been reported in patients receiving irbesartan:

- **Body as a Whole:** fever, chills, facial oedema, upper extremity oedema
- **Cardiovascular:** hypertension, cardiac murmur, myocardial infarction, angina pectoris, arrhythmic/conduction disorder, cardio-respiratory arrest, heart failure, hypertensive crisis, sinus abnormality, syncope
- **Dermatologic:** pruritus, dermatitis, ecchymosis, erythema face,
- **Endocrine/Metabolic/Electrolyte Imbalances:** gout
- **Gastrointestinal:** constipation, oral lesion, gastroenteritis, flatulence, abdominal distention
- **Musculoskeletal/Connective Tissue:** extremity swelling, arthritis, , joint stiffness, bursitis, muscle weakness
- **Nervous System:** sleep disturbance, numbness, somnolence, emotional disturbance, depression, paresthesia, tremor, transient ischemic attack, cerebrovascular accident, anxiety/nervousness
- **Renal/Genitourinary:** abnormal urination, prostate disorder, urinary tract infection
- **Respiratory:** epistaxis, tracheobronchitis, congestion, pulmonary congestion, dyspnea, wheezing, influenza, pharyngitis, rhinitis
- **Special Senses:** vision disturbance, hearing abnormality, ear infection, ear pain, conjunctivitis, other eye disturbance, eyelid abnormality, ear abnormality

Laboratory Test Findings

- Minor increases in blood urea nitrogen (BUN) have been reported in patients with essential hypertension receiving irbesartan.
- Mean decreases in neutropenia (<1000 cells/mm3) have been reported in patients receiving irbesartan

Rare cases of rhabdomyolysis have been reported in patients receiving angiotensin II receptor blockers.

USE IN SPECIAL POPULATIONS

Pregnancy

USE IN PREGNANCY
The use of Angiotensin II Receptor Antagonists (AIIRAs) is not recommended during the first trimester of pregnancy (see **WARNINGS AND PRECAUTIONS**). The use of AIIRAs is contraindicated during the second and third trimesters of pregnancy (see **CONTRAINDICATIONS AND WARNINGS AND PRECAUTIONS**).

The use of Angiotensin II Receptor Antagonists (AIIRAs) is not recommended during the first trimester of pregnancy. The use of AIIRAs is contraindicated during the second and third trimester of pregnancy.

Epidemiological evidence regarding the risk of teratogenicity following exposure to ACE inhibitors during the first trimester of pregnancy has not been conclusive; however a small increase in risk cannot be excluded. Whilst there is no controlled epidemiological information on the risk with Angiotensin II Receptor Antagonists (AIIRAs), similar risks may exist for this class of drugs. Unless continued AIIRAs therapy is considered essential, patients planning pregnancy should be changed to alternative anti-hypertensive treatments which have an established safety profile for use in pregnancy. When pregnancy is diagnosed, treatment with AIIRAs should be stopped immediately, and, if appropriate, alternative therapy should be started.

Exposure to AIIRAs therapy during the second and third trimesters is known to induce human fetotoxicity (decreased renal function, oligohydramnios, skull ossification retardation) and neonatal toxicity (renal failure, hypotension, hyperkalemia).

Should exposure to AIIRAs have occurred from the second trimester of pregnancy, ultrasound check of renal function and skull is recommended.

Infants whose mothers have taken AIIRAs should be closely observed for hypotension.

Lactation

Because no information is available regarding the use of irbesartan during breast-feeding, irbesartan is not recommended and alternative treatments with better established safety profiles during breast-feeding are preferable, especially while nursing a newborn or preterm infant (see **CONTRAINDICATIONS**).

Pediatrics

Safety and efficacy of irbesartan has not been established in children.

Geriatrics

No overall differences in effectiveness or safety has been reported between elderly subjects and younger subjects, but greater sensitivity of some older individuals cannot be ruled out.

OVERDOSE

Experience in adults exposed to doses of up to 900 mg/day for 8 weeks revealed no toxicity. The most likely manifestations of overdose are expected to be hypotension and tachycardia; bradycardia might also occur from overdose. No specific information is available on the treatment of overdose with irbesartan. The patient should be closely monitored, and the treatment should be symptomatic and supportive. Suggested measures include induction of emesis and/or gastric lavage. Activated charcoal may be useful in the treatment of overdose. Irbesartan is not removed by haemodialysis.

Storage: Store below 30°C

KEEP ALL MEDICINES OUT OF REACH OF CHILDREN

Supply: Blister pack of 2 x 14's

Information compiled in August, 2019

Manufactured in India by:
Sun Pharmaceutical Ind. Ltd.
Village Ganguwala, Poota Sahib- 173025
Distt. Simour, Himachal Pradesh (India).

Product Registration Holder:
RANBAXY (MALAYSIA) SDN. BHD.
(A Sun Pharma Company)
Lot 23, Bakar Arang Industrial Estate,
08000 Sungai Petani, Kedah,
MALAYSIA

Black

IRBEL TABLETS PIL
Size : 122 x 384 mm
Market : Malaysia
SPIL/PKGDEV - J11/DEC/2018-V01
J31/DEC/2018-V02, AK16/Aug/2019-V03

SAP CODE : 0000000
PHARMA CODE : 00000

Font Size : 6.4 pt

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